



# Greedy Algorithms 1

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# What is a Greedy Strategy?



- ✓ ● A strategy that assumes that the best answer can be found using some (not all) possibilities and only tries out those limited possibilities
- It involves coming up with a claim (greedy) and then proving it
- How to prove a Greedy Strategy?
  - Formal or intuitive proof
  - Trying out too many cases and failing to disprove

# What is a Greedy Strategy?



- A strategy that assumes that the best answer can be found using some (not all) possibilities and only tries out those limited possibilities
- It involves coming up with a claim (greedy) and then proving it
- How to prove a Greedy Strategy?
  - Formal or intuitive proof *math* *get feeling* *last 2 elements contain the max answer (after sorting)*
  - Trying out too many cases and failing to disprove

prove a claim

actual proving

formal

intuitive

trying out  
too many

counter  
cases  
such that  
none of  
them fail  
the claim

gut feeling / common  
sense

improve with experience

stress

testing

everyone understands  
and believes

①

you understand  
and believe

③

everyone understands

②

but  
only you believe

|    |   |   |   |   |   |   |    |
|----|---|---|---|---|---|---|----|
| 11 | 6 | 7 | 2 | 0 | 2 | 9 | 10 |
|----|---|---|---|---|---|---|----|

sorted

|   |   |   |   |   |   |    |    |
|---|---|---|---|---|---|----|----|
| 0 | 2 | 2 | 6 | 7 | 9 | 10 | 11 |
|---|---|---|---|---|---|----|----|

2 elements which have the max sum

for ( $i \rightarrow 0$  to  $n$ )

for ( $j \rightarrow i+1$  to  $n$ )

$\text{max\_val} = \max(\text{max\_val}, a[i] + a[j])$

$\neq$  trying all possibilities

after sorting  $\rightarrow$  how many possibilities to try

①



## Let's take some examples

- Given an array, find a subset of  $K$  elements with maximum sum
- Given an array, find the maximum difference between 2 elements
- Given an array, find the minimum difference between 2 elements
- Given an array, reorder the array so that sum of  $a[i] * i$  is maximized

→ of  $n$  elements

$$-10^9 \leq a(i) \leq 10^9$$

prob ①

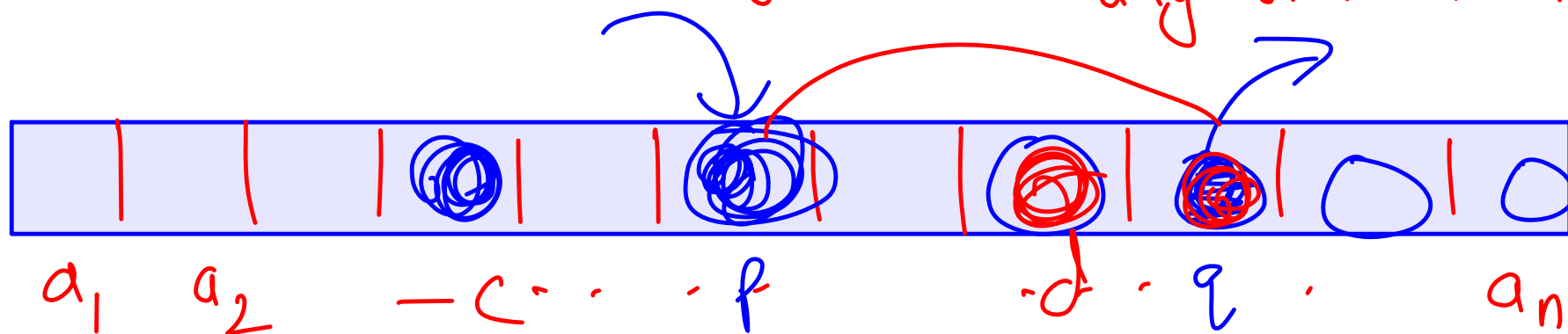
claim ① last  $k$  elements in sorted array

highest  $k$  elements in array

intuition

GA = Greedy answer

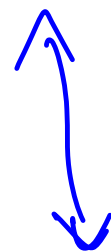
OA = any other answer



$$a_i \leq a_{i+1} \quad \forall i, 1 \leq i \leq n-1$$

$$GA = a_n + a_{n-1} + \dots + a_{n-k+1}$$

$$OA_i = a_n + a_{n-1} + \dots + a_{n-k+1} - a_q + a_p$$



$$GA - OA_1 = a_q - a_p$$

$$p < q$$

$$GA - OA_1 \geq 0$$

$$\underline{a_p \leq a_q}$$

$$a_q - a_p \geq 0$$

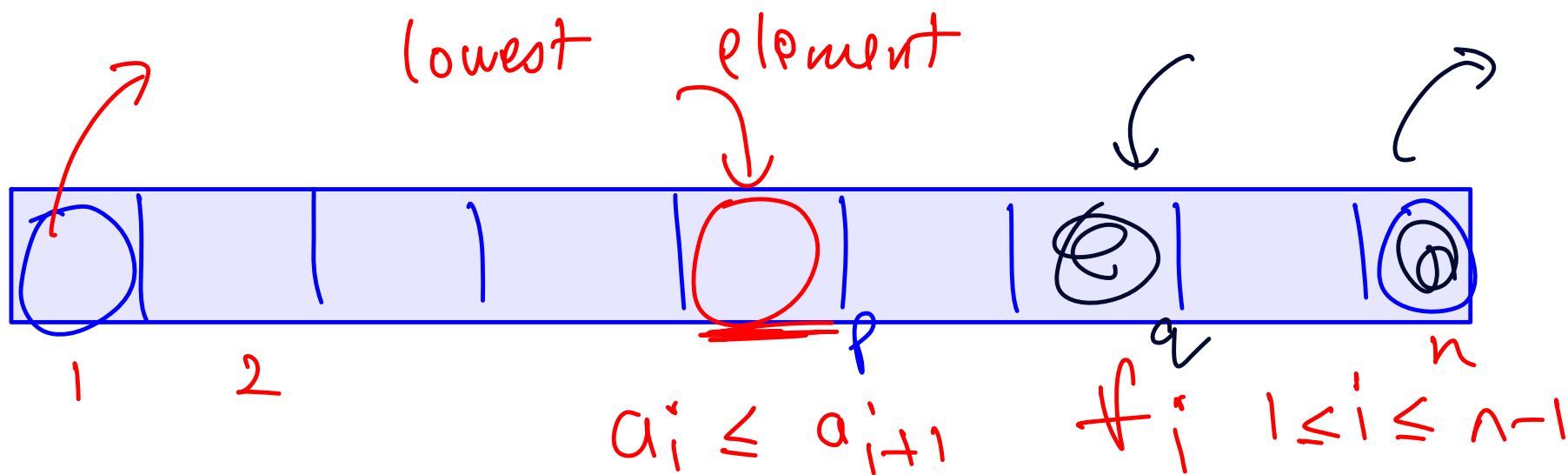
if I create any other answer  
 by swapping 1 element out of GA  
 and bringing some other element,

$$GA \geq OA_1$$

prob ②

max diff b/w any 2 elements

claim: max diff = diff b/w highest and lowest element



$$GA = a_n - a_1$$

$$OA = a_n - a_p$$

swap  $a_1$

$$GA - OA \geq 0$$

$$\underline{a_p - a_1} \geq 0 \quad \uparrow$$

$$GA = a_n - a_1$$

$$OA = a_q - a_1$$

swap  $a_n$

$$GA - OA \geq 0$$

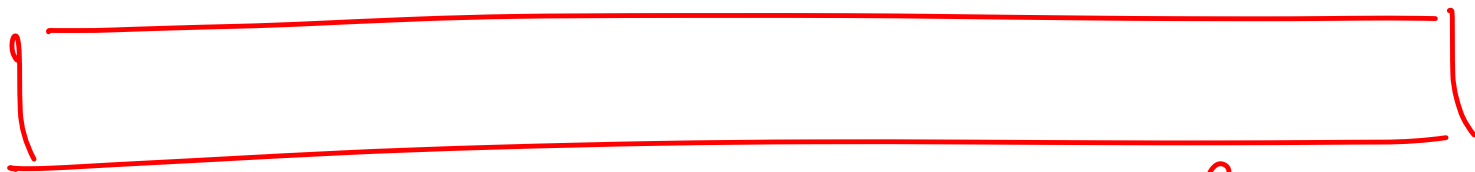
$$a_n - a_q \geq 0$$

Ins ③ min diff b/w any 2 elements

① claim  $\rightarrow$  sort and do  $a_{n/2+1} - a_{n/2}$

|   |   |    |    |
|---|---|----|----|
| 8 | 9 | 15 | 16 |
|---|---|----|----|

② claim  $\rightarrow$  try all adjacent elements in sorted array



$$a_i \leq a_{i+1} \quad \forall i, 1 \leq i \leq n-1$$

$$\min(a_2 - a_1, a_3 - a_2, \dots, a_n - a_{n-1})$$

$$GA = \min \{ \underbrace{a_2 - a_1}, \underbrace{a_3 - a_2}, \dots, a_n - a_{n-1} \}$$

$$\rightarrow \text{GAP } \underline{a_i - a_{i-1}}$$

any other pair  $\boxed{a_i}$   
 $a_j - a_j$   
 such that  $j < i-1$

$$\begin{aligned} \text{GAP} - \text{OAP} &= a_i - a_{i-1} - (a_i - a_j) \end{aligned}$$

$$\text{OAP } \underline{a_i - a_j}$$

$$= \underline{a_j - a_{i-1}} \leq 0$$

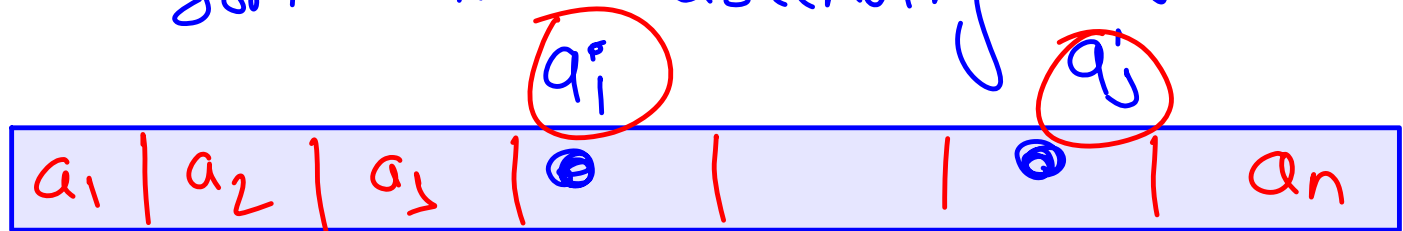
$$\underline{\text{GAP} \leq \text{OAP}}$$

prob (4) reorder the array so that  

$$\sum_{i=1}^n i \times a_i$$
 is maximised.

eg  $\rightarrow$  10 6 7 1

claim — sort in ascending order



$$a_i \leq a_{i+1} \quad \forall i$$

$$1 \leq i \leq n-1$$

$$GA = a_1 \times 1 + a_2 \times 2 + \dots + a_n \times n$$

$$OA = a_1 \times 1 + \dots + a_{i-1} \times (i-1) + a_j \times i + \dots + a_i \times j + \dots + a_n \times n$$

$a_i$  and  $a_j$

$$i < j \Rightarrow a_i \leq a_j$$

$$\underline{\underline{GA - OA}} = (a_i x_i + a_j x_j) - (a_j x_i + a_i x_j)$$

$$= a_i x (i - j) - a_j (i - j)$$

$$= \underline{\underline{(a_i - a_j)}} \underline{\underline{(i - j)}}$$

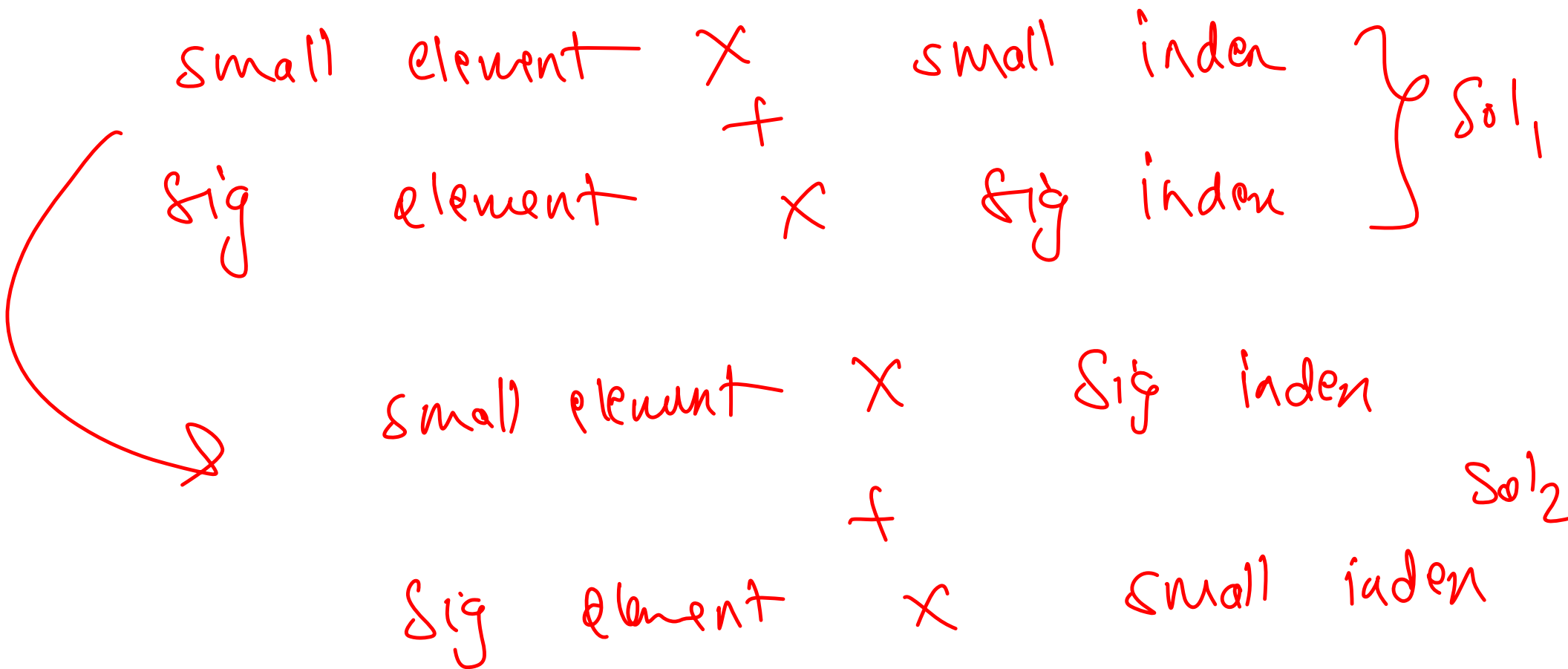
$$\leq 0 \times \leq 0$$

$$\underline{\underline{\geq 0}}$$

$$\Rightarrow GA - OA \geq 0$$

$$\underline{\underline{GA \geq OA}}$$

1 swap  $\rightarrow$  against the greedy strategy

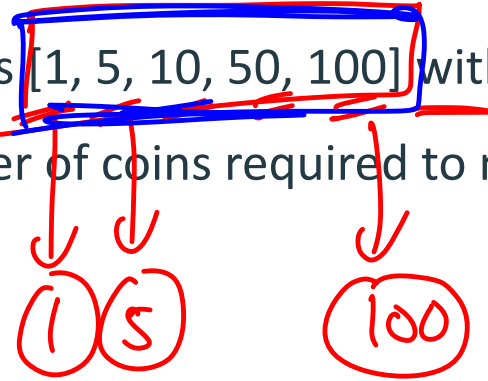


$Sol_1 \succcurlyeq Sol_2 \succcurlyeq Sol_3 \succcurlyeq Sol_4 \succcurlyeq Sol_5$

# Coin Change Problem: [Link](#) [raw link on last slide]



Given the following denominations of coins [1, 5, 10, 50, 100] with unlimited supply, find the minimum number of coins required to make a sum of  $X$  ( $1 \leq X \leq 1e9$ )



Examples:

X = 125 can be made with 100 + 10 + 10 + 5

X = 256 can be made with 100 + 100 + 50 + 5 + 1

Claim  $\rightarrow$  pick up highest den. possible and use  
it as many times as possible  
then go to the next highest  
and then to the next .....

$$152 \rightarrow 100 + 50 + 1 + 1$$

why does it work

$$X = 7 \times \textcircled{1} + 10 \times \textcircled{5} + 6 \times \textcircled{10} + 1 \times \textcircled{100}$$

$5 \times \textcircled{1}$  is worse than  
 $1 \times \textcircled{5}$

$$k \times \textcircled{1}$$



$$k-5 \times \textcircled{1}$$

$$+$$

$$1 \times \textcircled{5}$$

$$\underline{\underline{k \geq 5}}$$

$$k \times 5$$



$$k-2 \times \textcircled{5}$$

$$+$$

$$1 \times \textcircled{10}$$

$$k \geq 2$$

$$k \times 10$$



$$k-5 \times \textcircled{10}$$

$$+$$

$$1 \times \textcircled{50}$$

$$k \geq 5$$

|    |    |    |    |     |
|----|----|----|----|-----|
| 1  | 5  | 10 | 50 | 100 |
| ↓  | ↓  | ↓  | ↓  |     |
| <5 | <2 | <5 | <2 |     |

lower den.

higher den.

if it is possible to convert  $k$  coins  
of lower den. to 1 coin of higher  
den. we should do it.

1      5      10      50      100

$$a_i \% a_{i-1} = 0 \quad \forall i$$

high multiples of low  $\triangleright$  < small multiples of high  $\triangleright$

$$12 \times \textcircled{1} \rightarrow 1 \times \textcircled{10} + 2 \times \textcircled{1}$$

$$12 \times \textcircled{1} \rightarrow 2 \times \textcircled{5} + 2 \times \textcircled{1}$$

$$1 \times \textcircled{10} + 2 \times \textcircled{1}$$

$$5 \% 1 = 0$$

$$10 \% 5 = 0$$

$$50 \% 10 = 0$$

$$100 \% 50 = 0$$



# Coin Change Problem ( dynamic programming )



Does the greedy strategy always work?

Let's change the denominations to  $[1, 8, 10]$ . Construct  $X = 16$

Greedy strategy  $\rightarrow [10, 1, 1, 1, 1, 1, 1]$

Optimal strategy  $\rightarrow [8, 8]$

$$1 \times 10 + 6 \times 1$$

$$2 \times 8$$

# Coin Change Problem

f i



$$a_i \% a_{i-1} = 0$$

When does the greedy strategy work then?

If we can make  $a[i + 1]$  with  $a[i]$ , we should always first try to use  $a[i + 1]$  as many times as possible and only then use  $a[i]$ , when required coin change is lesser than  $a[i + 1]$

# Maximum Product Problem



Given  $N$  ( $1 \leq N \leq 1e9$ ), find two numbers  $A$  and  $B$  such that  $A + B = N$  and  $A * B$  is maximized.

(A and B have to be integers)

Example:

$N = 5$ , ( $1 * 4 = 4$ ), ( $2 * 3 = 6$ ), ( $3 * 2 = 6$ ), ( $4 * 1 = 4$ )

$N = 8$ , ( $1 * 7 = 7$ ), ( $2 * 6 = 12$ ), ( $3 * 5 = 15$ ), ( $4 * 4 = 16$ ), ( $5 * 3 = 15$ )...

claim :  $A = \text{floor}(N/2)$   
 $B = \text{ceil}(N/2)$

$N = 5$        $A = 2$  ,  $B = 3$

$N = 6$        $A = 3$  ,  $B = 3$

$GA = \overset{A}{\text{floor}}(N/2) \times \overset{B}{\text{ceil}}(N/2)$

$A = 5$  ,  $B = 2 \Rightarrow A = 2$  ,  $B = 5$

only look at pairs where  $\boxed{A \leq B}$

$OA = (\text{floor}(N/2) - k) \cdot (\text{ceil}(N/2) + k)$   
 $\underbrace{\hspace{10em}}_{k > 0}$

$$\underline{OA} = \underline{f(N|_2) \cdot k} + \cancel{f(N|_2) \cdot c(N|_2)} - c(N|_2) \cdot k - k^2$$

$$\underline{GA - OA} = \cancel{f(N|_2) \cdot c(N|_2)} - \left[ \cancel{f(N|_2) \cdot c(N|_2)} \right]$$

$$= -f(N|_2) \cdot k + c(N|_2) \cdot k + k^2$$

$$= k^2 + k ( \underline{c(N|_2) - f(N|_2)} )$$

$$\stackrel{11}{=} +ve + +ve \times ( \geq 0 )$$

$$= +ve + \geq 0 \Rightarrow \underline{+ve}$$

# Maximum Product Problem



Can we observe a pattern?

N = 12

(1 \* 11 = 11), (2 \* 10 = 20), (3 \* 9 = 27), (4 \* 8 = 32), (5 \* 7 = 35), (6 \* 6 = 36),  
(7 \* 5 = 35), (8 \* 4 = 32), (9 \* 3 = 27), (10 \* 2 = 20), (11 \* 1 = 11)

How to formally prove this?



# Minimum Dot Product

Given 2 vectors A [a1, a2, a3, ... vn] and B [b1, b2, b3, ... bn], rearrange the elements in the vectors so that the dot product is minimized.

Example:

$$\sum_{i=1}^n a_i \cdot b_i$$

$$A = [-1, 3, -2], B = [-10, 1, 5] \Rightarrow [-1 * -10 + 3 * 1 + -2 * 5] = [10 + 3 + -10] = 3$$

$$A = [-2, -1, 3], B = [5, 1, -10] \Rightarrow [-2 * 5 + -1 * 1 + -10 * 3] = [-10 + -1 + -30] = -41$$

Claim: It we keep both sorted in ascending

$$\rightarrow a_1 \times b_n + a_2 \times b_{n-1} + a_3 \times b_{n-2} \dots a_n \times b_1$$
$$= \sum_{i=1}^n a_i \times b_{n-i+1}$$

sort 1 in ascending and the other in descending

a smaller number from A with a larger  
no from B

and  
higher number from A with a smaller  
no from B

$$a_1 \ a_2 \ a_3 \ a_i \ \dots \ a_j \ \dots \ a_n$$

$$a_i \leq a_{i+1}$$

$$b_1 \ b_2 \ b_3 \ \dots \ b_n$$

$$b_i \geq b_{i+1}$$

$$GA = a_1 \times b_1 + a_2 \times b_2 + \dots + a_n \times b_n$$

$$OA = a_1 \times b_1 + \dots + a_{i-1} \times b_{i-1} + a_j \times b_i + \dots$$

$$+ \dots + a_i \times b_j + \dots + a_n \times b_n$$

$$\boxed{S_A \times B_B \text{ and } B_A \times S_B}$$

$$\hookrightarrow \boxed{B_A \times B_B \text{ and } S_A \times S_B}$$

$$GA - 0A = a_i \times b_i + a_j \times b_j$$

$$- a_j \times b_i - a_i \times b_j$$

$$= a_i(b_j - b_i) - a_j(b_i - b_j)$$

$$= \underbrace{(a_i - a_j)}_{\leq 0} \times \underbrace{(b_i - b_j)}_{\geq 0}$$

$$= \leq 0 \quad i < j$$

$$= \leq 0 \quad \Downarrow$$

$$a_i \leq a_j$$

$$b_j \geq b_i$$

$$GA - 0A \leq 0$$

$$\underline{\underline{GA \leq 0A}}$$



# Minimum Dot Product

Can we try out for 2 elements? Assuming we have 2 sorted vectors:

$$A = [a_1, a_2] \quad a_2 > a_1 \quad | \quad B = [b_1, b_2] \quad \underline{\underline{b_2 > b_1}}$$

$$X = a_1 * b_1 + a_2 * b_2 \quad | \quad Y = a_1 * b_2 + a_2 * b_1$$

$$X = Y + (a_1 * b_1 - a_2 * b_2 + a_2 * b_2 - a_2 * b_1)$$

$$X = Y + (b_2 - b_1) * (a_1 - a_2)$$

$$a_1 \times (b_2 - b_1) \\ - a_2 \times (b_2 - b_1)$$

Since  $b_2 - b_1 \geq 0$  &  $a_1 - a_2 \leq 0$ , this means that  $X \leq Y$

# Minimum Dot Product



Shouldn't we try to multiple bigger numbers in A with smaller numbers in B  
and smaller numbers in A with bigger numbers in B to get the most optimal  
answer?

But how can we prove that this actually works?



# Minimum Dot Product

$$A = a_1 \leq a_2 \leq a_3 \dots \leq a_{n-2} \leq a_{n-1} \leq a_n$$

$$B = b_1 \geq b_2 \geq b_3 \dots \geq b_{n-2} \geq b_{n-1} \geq b_n$$

What if we swap 2 adjacent elements?

$$A = a_1 \ a_2 \ a_3 \ \dots \ a_k \ a_{k+1} \ \dots \ a_{n-2} \ a_{n-1} \ a_n$$

$$B = b_1 \ b_2 \ b_3 \ \dots \ b_{k+1} \ b_k \ \dots \ a_{n-2} \ a_{n-1} \ a_n$$

What is the change in the total sum  $\Rightarrow [a_k - a_{k+1}] * [b_{k+1} - b_k]$  which is  $\geq 0$



# Problem Links

- Minimum Number Of Coins:

[https://www.geeksforgeeks.org/problems/-minimum-number-of-coins4426/1?itm\\_source=geeksforgeeks&itm\\_medium=article&itm\\_campaign=bottom\\_sticky\\_on\\_article](https://www.geeksforgeeks.org/problems/-minimum-number-of-coins4426/1?itm_source=geeksforgeeks&itm_medium=article&itm_campaign=bottom_sticky_on_article)